Shun Watanabe - Publication List

Updated from the supplied publication-list PDF. Peer-reviewed papers are numbered chronologically: oldest = No. 1; newest entries appear first.

Selected Publications

Selected recent highlights and representative Nature / Science-family publications. Numbering follows the full chronological publication list.

98. Hai Wang, Kui Feng, Takahiro Kaneta, Tomoki Furukawa, Xuan Guo, Xinrui Wang, Jun Takeya, and Shun Watanabe *Coherent Charge Transport Enhanced by Programmed Electrochemical Doping in Conjugated Polymers*. Small Methods, e70696 (2026).

93. Tomoki Furukawa, Naotaka Kasuya, Hiroshi Takayanagi, Shun Watanabe, and Jun Takeya *Hall mobility exceeding 100 cm^2 V^{-1} s^{-1} observed in strained organic semiconductors*. Science Advances, 11, eaea1634 (2025).

92. Nobutaka Osakabe, Jeongeun Her, Takahiro Kaneta, Ayano Tajima, Elena Longhi, Kaixiang Tang, Kazuto Fujimori, Stephen Barlow, Seth R. Marder, Shun Watanabe, Jun Takeya, and Yu Yamashita *Polymeric microwave rectifiers enabled by monolayer-thick ionized donors*. Science Advances, 11, eadv9952 (2025).

87. Yu Yamashita, Shinya Kohno, Elena Longhi, Samik Jhulki, Shohei Kumagai, Stephen Barlow, Seth R. Marder, Jun Takeya, and Shun Watanabe* (*corresponding author) *N-type molecular doping of a semicrystalline conjugated polymer through cation exchange*. Communications Materials 5 79 (2024).

82. Masaki Ishii, Yu Yamashita, Shun Watanabe, Katsuhiko Ariga, and Jun Takeya *Doping of molecular semiconductors through proton-coupled electron transfer*. Nature 622 285 (2023).

71. Naotaka Kasuya, Junto Tsurumi, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Two-dimensional hole gas in organic semiconductors*. Nature Materials 20 1401 (2021).

61. Taiki Sawada, Akifumi Yamamura, Mari Sasaki, Kayo Takahira, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Correlation between the static and dynamic responses of organic single-crystal field-effect transistors*. Nature Communications, 11, 4849 (2020).

31. Junto Tsurumi, Hiroyuki Matsui, Takayoshi Kubo, Roger Hausermann, Chikahiko Mitsui, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Coexistence of ultra-long spin relaxation time and coherent charge transport in organic single-crystal semiconductors*. Nature Physics, 13, 994 (2017).

21. Keehoon Kang*, Shun Watanabe*, Katharina Broch, Alessandro Sepe, Adam Brown, Iyad Nasrallah, Mark Nikolka, Zhuping Fei, Martin Heeney, Daisuke Matsumoto, Kazuhiro Marumoto, Hisaaki Tanaka, Shin-ichi Kuroda, and Henning Sirringhaus (*equal contribution) *2D coherent charge transport in highly ordered conducting polymers doped by solid state diffusion*. Nature Materials, 15, 896 (2016).

17. Shun Watanabe*, Kazuya Ando*, Keehoon Kang, Sebastian Mooser, Yana Vaynzof, Hidekazu Kurebayashi, Eiji Saitoh, and Henning Sirringhaus (*equal contribution) *Polaron spin current transport in organic semiconductors*. Nature Physics, 10, 308 (2014).

14. Kazuya Ando*, Shun Watanabe*, Sebastian Mooser, Eiji Saitoh, and Henning Sirringhaus (*equal contribution) *Solution-processed organic spin-charge converter*. Nature Materials, 12, 622 (2013).

Full Publication List

98 peer-reviewed papers. Newest entries are shown first; numbering is chronological from the oldest paper.

98. Hai Wang, Kui Feng, Takahiro Kaneta, Tomoki Furukawa, Xuan Guo, Xinrui Wang, Jun Takeya, and Shun Watanabe *Coherent Charge Transport Enhanced by Programmed Electrochemical Doping in Conjugated Polymers*. Small Methods, e70696 (2026).

97. Yoshihisa Usami, Yu Yamashita, Tomohiro Murata, Takafumi Matsumoto, Masataka Ito, Shun Watanabe, and Jun Takeya *Scalable fabrication of precise flexible strain sensors using organic semiconductor single crystals*. Science and Technology of Advanced Materials, 26, 2451020 (2025).

96. Mizuki Abe, Yu Yamashita, Taiki Sawada, Tatsuyuki Makita, Shohei Kumagai, Shun Watanabe, and Jun Takeya *Strained Organic Thin-Film Single Crystals for High-Mobility and High-Frequency Transistors*. Advanced Electronic Materials, 11, 2500144 (2025).

95. Aoi Morimoto, Yu Yamashita, Taiki Sawada, Masaki Ishii, Shun Watanabe, and Jun Takeya *Doped interlayers enabling high-mobility p-type organic transistors with copper contact electrodes*. Journal of Materials Chemistry C, 13, 17544-17550 (2025).

94. Nobutaka Osakabe, Akifumi Yamamura, Tatsuyuki Makita, Yasunari Tamai, Shun Watanabe, and Jun Takeya *Extraordinarily persistent photo-induced carriers mediated by interfacial exciton splitter in organic single-crystalline semiconductors*. Applied Physics Express, 18, 051001 (2025).

93. Tomoki Furukawa, Naotaka Kasuya, Hiroshi Takayanagi, Shun Watanabe, and Jun Takeya *Hall mobility exceeding 100 cm^2 V^{-1} s^{-1} observed in strained organic semiconductors*. Science Advances, 11, eaea1634 (2025).

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91. Hai Wang, Gaiqing Zhao, Weimin Li, Yue Wang, Tao Yang, Shun Watanabe, and Xiaobo Wang *Uniform Zn Stripping/Plating Enable by PANI:PSS/Cellulose Framework for Dendrite-Free Rechargeable Aqueous Zinc-Ion Batteries*. Journal of Physical Chemistry C, 128, 13722 (2024).

90. Hai Wang, Qin Zhao, Yue Wang, Junliang Lin, Weimin Li, Shun Watanabe, and Xiaobo Wang *An anti-corrosive cellulose nanocrystal/carbon nanotube derived Zn anode interface for dendrite-free aqueous Zn-ion batteries*. Physical Chemistry Chemical Physics, 26, 23411 (2024).

89. Hai Wang, Qin Zhao, Weimin Li, Shun Watanabe, and Xiaobo Wang *A dendrite-free Zn anode enabled by PEDOT:PSS/MoS_2 electrokinetic channels for aqueous Zn-ion batteries*. Nanoscale, 16, 7200-7210 (2024).

88. Yu Yamashita, Harumi Hayakawa, Pushi Wang, Tatsuyuki Makita, Shohei Kumagai, Shun Watanabe, and Jun Takeya *Ion sensors based on organic semiconductors acting as quasi-reference electrodes*. Proceedings of the National Academy of Sciences, 121, e2405933121 (2024).

87. Yu Yamashita, Shinya Kohno, Elena Longhi, Samik Jhulki, Shohei Kumagai, Stephen Barlow, Seth R. Marder, Jun Takeya, and Shun Watanabe* (*corresponding author) *N-type molecular doping of a semicrystalline conjugated polymer through cation exchange*. Communications Materials 5 79 (2024).

86. Kazuyoshi Watanabe, Naoki Miura, Hiroaki Taguchi, Takeshi Komatsu, Atsushi Aratake, Tatsuyuki Makita, Masahiro Tanabe, Takahiro Wakimoto, Shohei Kumagai, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *All-Carbon-Based Complementary Integrated Circuits*. Advanced Materials Technologies, 9, 2301673 (2024).

85. Zijing Guo, Tetsu Sato, Yang Han, Naoki Takamura, Ryohei Ikeda, Tatsuya Miyamoto, Noriaki Kida, Makiko Ogino, Youtaro Takahashi, Naotaka Kasuya, Shun Watanabe, Jun Takeya, Qingshuo Wei, Masakazu Mukaida, and Hiroshi Okamoto *Band transport evidence in PEDOT:PSS films using broadband optical spectroscopy from terahertz to ultraviolet region*. Communications Materials 5 26 (2024).

84. Naoyuki Niitsu, Masato Mitani, Hiroyuki Ishii, Nobuhiko Kobayashi, Kenji Hirose, Shun Watanabe, Toshihiro Okamoto, and Jun Takeya *Powder x-ray diffraction analysis with machine learning for organic-semiconductor crystal-structure determination*. Applied Physics Letters, 125, 1 (2024).

83. Xiaozhu Wei, Shohei Kumagai, Tatsuyuki Makita, Kotaro Tsuzuku, Akifumi Yamamura, Mari Sasaki, Shun Watanabe*, and Jun Takeya* (*corresponding author) *High-speed hybrid complementary ring oscillators based on solution-processed organic and amorphous metal oxide semiconductors*. Communications Materials 4 4 (2023).

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81. Tadanori Kurosawa, Yu Yamashita, Yuki Kobayashi, Craig P. Yu, Shohei Kumagai, Tsubasa Mikie, Itaru Osaka, Shun Watanabe, Jun Takeya, and Toshihiro Okamoto *Employment of Alkoxy Sidechains in Semicrystalline Semiconducting Polymers for Ambient-Stable p-Doped Conjugated Polymers*. Macromolecules 57 328 (2023).

80. Taehyun Won, Shohei Kumagai, Naotaka Kasuya, Yu Yamashita, Shun Watanabe, Toshihiro Okamoto, and Jun Takeya *Individual and synergetic charge transport properties at the solid and electrolyte interfaces of a single ultrathin single crystal of organic semiconductors*. Physical Chemistry Chemical Physics, 25, 14496 (2023).

79. Y. Han, T. Miyamoto, T. Otaki, N. Takamura, N. Kida, N. Osakabe, J. Tsurumi, S. Watanabe, T. Okamoto, J. Takeya, and H. Okamoto *Scattering mechanism of hole carriers in organic molecular semiconductors deduced from analyses of terahertz absorption spectra using Drude-Anderson model*. Applied Physics Letters, 120, 5 (2023).

78. Akchheta Karki, Yu Yamashita, Shangzhi Chen, Tadanori Kurosawa, Jun Takeya, Vallery Stanishev, Vanya Darakchieva, Shun Watanabe, and Magnus P. Jonsson *Doped semiconducting polymer nanoantennas for tunable organic plasmonics*. Communications Materials 3 48 (2022).

77. Shohei Kumagai, Tatsuyuki Makita, Shun Watanabe*, and Jun Takeya* (*corresponding author) (Review) *Scalable printing of two-dimensional single crystals of organic semiconductors towards high-end device applications*. Applied Physics Express 15 030101 (2022).

76. Kazuyoshi Watanabe, Naoki Miura, Hiroaki Taguchi, Takeshi Komatsu, Hideyuki Nosaka, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Electrostatically-sprayed carbon electrodes for high performance organic complementary circuits*. Scientific Reports 12 16009 (2022).

75. Kazuyoshi Watanabe, Naoki Miura, Hiroaki Taguchi, Takeshi Komatsu, Hideyuki Nosaka, Toshihiro Okamoto, Yu Yamashita, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Improvement of contact resistance at carbon electrode/organic semiconductor interfaces through chemical doping*. Applied Physics Express 15 101005 (2022).

74. Kenta Takada, Kohei Yoshimi, Satoshi Tsutsui, Koji Kimura, Kouichi Hayashi, Ikutaro Hamada, Susumu Yanagisawa, Naotaka Kasuya, Shun Watanabe, Jun Takeya, and Yusuke Wakabayashi *Phonon dispersion of the organic semiconductor rubrene*. Physical Review B, 105, 205205 (2022).

73. Yu Yamashita, Samik Jhulki, Dinesh Bhardwaj, Elena Longhi, Shohei Kumagai, Shun Watanabe, Stephen Barlow, Seth R. Marder and Jun Takeya *Highly air-stable, n-doped conjugated polymers achieved by dimeric organometallic dopants*. Journal of Materials Chemistry C, 12, 4105 (2022).

72. Hiroyuki Ishii, Naotaka Kasuya, Nobuhiko Kobayashi, Kenji Hirose, Shohei Kumagai, Shun Watanabe, and Jun Takeya *Gate induced modulation of electronic states in monolayer organic field-effect transistor*. . Applied Physics Letters, 119, 22 (2022).

71. Naotaka Kasuya, Junto Tsurumi, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Two-dimensional hole gas in organic semiconductors*. Nature Materials 20 1401 (2021).

70. Masato Ito, Yu Yamashita, Taizo Mori, Masaaki Chiba, Takayuki Futae, Jun Takeya, Shun Watanabe*, and Katsuhiko Ariga* (*corresponding author) *Hyper 100 degC Langmuir-Blodgett (Langmuir-Schaefer) technique for organized ultrathin film of polymeric semiconductors*. Langmuir 38 5237 (2021).

69. Masato Ito, Yu Yamashita, Taizo Mori, Katsuhiko Ariga, Jun Takeya, and Shun Watanabe* (*corresponding author) *Band mobility exceeding 10 cm^2 V^{-1} s^{-1} assessed by field-effect and chemical double doping in semicrystalline polymeric semiconductors*. Applied Physics Letters 119 1 (2021).

68. Tadanori Kurosawa, Toshihiro Okamoto, Yu Yamashita, Shohei Kumagai, Shun Watanabe, and Jun Takeya *Strong and Atmospherically Stable Dicationic Oxidative Dopant*. Advanced Science 8 2101998 (2021).

67. Xiaozhu Wei, Shohei Kumagai, Mari Sasaki, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Stabilizing solution-processed metal oxide thin-film transistors via trilayer organic-inorganic hybrid passivation*. AIP Advances, 11, 035309 (2021).

66. Tatsuyuki Makita, Yoma Ninomiya, Shohei Kumagai, Toshihiro Okamoto, Mari Sasaki, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Nano-Ground Glass as a Superhydrophilic Template for Printing High-Performance Organic Single -Crystal Thin Films*. Advanced Materials Interfaces 8 2100033 (2021).

65. Yu Yamashita, Junto Tsurumi, Tadanori Kurosawa, Kan Ueji, Yukina Tsuneda, Shinya Kohno, Hideto Kempe, Shohei Kumagai, Toshihiro Okamoto, Jun Takeya, and Shun Watanabe* (*corresponding author) *Supramolecular cocrystals built through redox-triggered ion intercalation in π-conjugated polymers*. Communications Materials, 2, 45 (2021).

64. Shun Watanabe*, Ryohei Hakamatani, Keita Yaegashi, Yu Yamashita, Han Nozawa, Mari Sasaki, Shohei Kumagai, Toshihiro Okamoto, Cindy G. Tang, Lay-Lay Chua, Peter K. H. Ho, and Jun Takeya (*corresponding author) *Surface Doping of Organic Single-Crystal Semiconductors to Produce Strain-Sensitive Conductive Nanosheets*. Advanced Science, 8, 2002065 (2021).

63. Takaaki Takenaka, Kota Ishihara, Masaki Roppongi, Yijie Miao, Yuta Mizukami, Tatsuyuki Makita, Junto Tsurumi, Shun Watanabe, Jun Takeya, Minoru Yamashita, Kiyoshi Torizuka, Yoshiya Uwatoko, Takahiko Sasaki, Xing Huang, Wei Xu, Daoben Zhu, Na Su, Jinguang Cheng, Takasada Shibauchi, and Kenichiro Hashimoto *Strongly correlated superconductivity in a copper-based metal-organic framework with a perfect kagome lattice*. Science Advances 7, 12 (2021).

62. Shinya Kohno, Yu Yamashita, Naotaka Kasuya, Tsubasa Mikie, Itaru Osaka, Kazuo Takimiya, Jun Takeya, and Shun Watanabe* (*corresponding author) *Controlled steric selectivity in molecular doping towards closest-packed supramolecular conductors*. Communications Materials, 1, 79 (2020).

61. Taiki Sawada, Akifumi Yamamura, Mari Sasaki, Kayo Takahira, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Correlation between the static and dynamic responses of organic single-crystal field-effect transistors*. Nature Communications, 11, 4849 (2020).

60. Masato Ito, Yu Yamashita, Yukina Tsuneda, Taizo Mori, Jun Takeya, Shun Watanabe*, and Katsuhiko Ariga* (*corresponding author) *100° C-Langmuir-Blodgett Method for Fabricating Highly Oriented, Ultrathin Films of Polymeric Semiconductors*. ACS Applied Materials and Interfaces, 12, 56522 (2020).

59. Shohei Kumagai, Shun Watanabe, Hiroyuki Ishii, Nobuaki Isahaya, Akifumi Yamamura, Takahiro Wakimoto, Hiroyasu Sato, Akihito Yamano, Toshihiro Okamoto, and Jun Takeya *Coherent Electron Transport in Air-Stable, Printed Single-Crystal Organic Semiconductor and Application to Megahertz Transistors*. Advanced Materials, 32, 2003245 (2020).

58. Tatsuyuki Makita, Ryohei Nakamura, Mari Sasaki, Shohei Kumagai, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Electroless-Plated Gold Contacts for High-Performance, Low Contact Resistance Organic Thin Film Transistors*. Advanced Functional Materials, 30, 2003977 (2020).

57. Tatsuyuki Makita, Akifumi Yamamura, Junto Tsurumi, Shohei Kumagai, Tadanori Kurosawa, Toshihiro Okamoto, Mari Sasaki, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Damage-free Metal Electrode Transfer to Monolayer Organic Single Crystalline Thin Films*. Scientific Reports, 10, 4702 (2020).

56. Akifumi Yamamura, Hiromasa Fujii, Hirohito Ogasawara, Dennis Nordlund, Osamu Takahashi, Yutaro Kishi, Hiroyuki Ishii, Nobuhiko Kobayashi, Naoyuki Niitsu, Balthasar Blülle, Toshihiro Okamoto, Yusuke Wakabayashi*, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Sub-molecular structural relaxation at a physisorbed interface with monolayer organic single-crystal semiconductors*. Communications Physics, 3, 20 (2020).

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54. Tatsuyuki Makita, Shohei Kumagai, Akihito Kumamoto, Masato Mitani, Junto Tsurumi, Ryohei Hakamatani, Mari Sasaki, Toshihiro Okamoto, Yuichi Ikuhara, Shun Watanabe*, and Jun Takeya* (*corresponding author) *High-performance, semiconducting membrane composed of ultrathin, single-crystal organic semiconductors*. Proceedings of the National Academy of Sciences, 117, 80 (2020).

53. Angela Wittmann, Guillaume Schweicher, Katharina Broch, Jiri Novak, Vincent Lami, David Cornil, Erik R. McNellis, Olga Zadvorna, Deepak Venkateshvaran, Kazuo Takimiya, Yves H. Geerts, Jérôme Cornil, Yana Vaynzof, Jairo Sinova, Shun Watanabe, and Henning Sirringhaus *Tuning spin current injection at ferromagnet-nonmagnet interfaces by molecular design*. Physical Review Letters, 124, 027204 (2020).

52. Xiaozhu Wei, Shohei Kumagai*, Kotaro Tsuzuku, Akifumi Yamamura, Tatsuyuki Makita, Mari Sasaki, Shun Watanabe*, and Jun Takeya (*corresponding author) *Solution-processed flexible metal-oxide thin-film transistors operating beyond 20MHz*. Flexible and Printed Electronics, 5, 015003 (2020).

51. Hyun Ho Choi, Hee Taek Yi, Junto Tsurumi, Jae Joon Kim, Alejandro L. Briseno, Shun Watanabe, Jun Takeya, Kilwon Cho, and Vitaly Podzorov *A large anisotropic enhancement of the charge* *carrier mobility of flexible organic transistors with strain: a Hall effect and Raman study*. A, 7(1), 1901824. Advanced Science 7, 1901824 (2020).

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49. Taiki Sawada, Tatsuyuki Makita, Akifumi Yamamura, Mari Sasaki, Yasunari Yoshimura, Teruaki Hayakawa, Toshihiro Okamoto, Shun Watanabe, Shohei Kumagai, and Jun Takeya *Low-voltage complementary inverters using solution-processed, high-mobility organic single-crystal transistors fabricated by polymer-blend printing*. Applied Physics Letters, 117, 033301 (2020).

48. Takashi Ohtaki, Tsubasa Terashige, Junto Tsurumi, Tatsuya Miyamoto, Noriaki Kida, Shun Watanabe, Toshihiro Okamoto, Jun Takeya, and Hiroshi Okamoto *Evaluations of nonlocal electron-phonon couplings in tetracene, rubrene, and C 10 -DNBDT-NW based on density functional theory*. Physical Review B, 102, 245201 (2020).

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46. Yu Yamashita, Junto Tsurumi, Masahiro Ohno, Ryo Fujimoto, Shohei Kumagai, Tadanori Kurosawa, Toshihiro Okamoto, Jun Takeya, and Shun Watanabe* (*corresponding author) *Efficient molecular doping of polymeric semiconductors driven by anion exchange*. Nature, 572, 634 (2019).

45. Shun Watanabe*, Masahiro Ohno, Yu Yamashita, Tsubasa Terashige, Hiroshi Okamoto, and Jun Takeya (*corresponding author) *Validity of the Mott formula and the origin of thermopower in π-conjugated semicrystalline polymers*. Physical Review B (Rapid Communications), 100, 241201(R) (2019).

44. Shohei Kumagai*, Akifumi Yamamura, Tatsuyuki Makita, Junto Tsurumi, Ying Ying Lim, Takahiro Wakimoto, Nobuaki Isahaya, Han Nozawa, Kayoko Sato, Masato Mitani, Toshihiro Okamoto, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Scalable Fabrication of Organic Single-Crystalline Wafers for Reproducible TFT Arrays*. Scientific Reports, 9, 15897 (2019).

43. Artem G. Shulga, Akifumi Yamamura, Kotaro Tsuzuku, Ryan M. Dragoman, Dmitry N. Dirin, Shun Watanabe, Maksym V. Kovalenko, Jun Takeya, and Maria A. Loi *Patterned Quantum Dot Photosensitive FETs for Medium Frequency Optoelectronics*. Advanced Materials Technologies, 4, 1900054 (2019).

42. Katsuhiko Ariga, Tatsuyuki Makita, Masato Ito, Taizo Mori, Shun Watanabe, and Jun Takeya *Review of advanced sensor devices employing nanoarchitectonics concepts*. (Review article) Beilstein Journal of Nanotechnology, 10, 2014 (2019).

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36. Shun Watanabe*, Hirotaka Sugawara, Roger Häusermann, Balthasar Blülle, Akifumi Yamamura, Toshihiro Okamoto, and Jun Takeya* (*corresponding author) *Remarkably low flicker noise in solution-processed organic single crystal transistors*. Communications Physics, 1, 37 (2018).

35. Akifumi Yamamura, Shun Watanabe, Mayumi Uno, Masato Mitani, Chikahiko Mitsui, Junto Tsurumi, Nobuaki Isahaya, Yusuke Kanaoka,Toshihiro Okamoto, and Jun Takeya *Wafer-scale,* *layer-controlled organic single crystals for high-speed circuit operation*. Science Advances, 4, eaao5758 (2018).

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32. Ryo Fujimoto, Yu Yamashita, Shohei Kumagai, Junto Tsurumi, Alexander Hinderhofer, Katharina Broch, Frank Schreiber, Shun Watanabe*, and Jun Takeya* (*corresponding author) *Molecular doping in organic semiconductors: fully solution-processed, vacuum-free doping with metal-organic complexes in an orthogonal solvent*. Journal of Materials Chemistry C, 5, 12023 (2017).

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25. Mohamad Insan Nugraha, Hiroyuki Matsui, Shun Watanabe, Takayoshi Kubo, Roger Hausermann, Satria Zulkarnaen Bisri, Mykhailo Sytnyk, Wolfgang Heiss, Maria Antonietta Loi, and Jun Takeya *Strain-Modulated Charge Transport in Flexible PbS Nanocrystal Field-Effect Transistors*. Advanced Electronic Materials, 3, 1600360 (2017).

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